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Lay beliefs about AI assessment of interpersonal skills in personnel selection

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Artificial intelligence (AI) is revolutionizing talent acquisition, with organizations increasingly replacing humans with AI technologies in personnel selection processes. Field studies of nurses provide preliminary evidence that AI can outperform humans in predicting interpersonal skills. However, a series of surveys and experiments document widespread lay beliefs in the opposite direction: individuals perceive AI as being less capable than humans in assessing interpersonal skills. This effect was not observed for analytical skills, suggesting that the results stem from lay beliefs about the suitability of AI in interpersonal contexts and not from a generalized aversion to algorithms. These lay beliefs bias managers' perceptions of employees: managers were less likely to assign tasks requiring interpersonal skills to AI-selected (vs. human-selected) employees. Additionally, applicants reported reducing their emphasis on interpersonal (vs. analytical) skills during AI (vs. human) selection processes. Informing participants about advanced AI selection technologies reduced negative lay beliefs about AI performance in the interpersonal domains.

Recruiting the best possible talent is a top priority for organizations¹. Artificial intelligence (AI) technology is transforming personnel selection. AI assesses applicants' strengths by analyzing their resumes, psychometric tests, chat and video interviews, and mini-game performance^{2,3}. Many organizations have already adopted these AI selection technologies at scale and this number is projected to increase substantially in the coming years^{2,3}. However, much about how individuals react to this new technology remains underexplored⁴. Addressing this gap is important given the profound implications such perceptions have on individuals' career paths and workplace dynamics. For example, if people hold negative perceptions about the specific skills of AI-selected employees, managers may treat them unfairly (e.g., in task assignments). Similarly, prospective candidates may underemphasize specific skills during the AI selection processes, potentially harming their job prospects when the lay beliefs are incorrect. Human resources (HR) professionals may also question the reliability of AI in identifying candidates' suitability for the required roles, losing confidence and opting not to use these services. Thus, for the successful integration of AI into personnel selection, understanding individuals' perceptions of this technology is essential. This paper explores lay beliefs about AI capabilities for personnel selection and their subsequent impacts. Lay beliefs refer to individuals' subjective and informal explanations or perceptions about the world around them⁵. Such beliefs may or may not be aligned with the scientific consensus. In our research, we focus on individuals' beliefs about the capabilities of AI-based selection tools (e.g., screening and interviewing technologies) used to evaluate job candidates^{6–8}.

We examine lay beliefs about AI capabilities and their consequences. We show that although our Field Study provides preliminary evidence that AI can effectively assess interpersonal skills under certain conditions, often surpassing human experts, people perceive AI (vs. humans) as less capable in the assessment of interpersonal (vs. analytical) skills. We further show that these lay beliefs can bias managers' perceptions of employees selected via AI interviews and influence interviewees' strategies in AI interviews. Finally, we suggest that informing people about advanced AI selection technologies can reduce negative lay beliefs about AI's capability to assess interpersonal skills.

Research has long documented an inherent resistance to quantified methods for personnel selection, which has extended to general negative views towards AI selection technologies (see Supplementary Information (SI) 1 for details). Our research, however, takes a more nuanced approach by investigating whether lay beliefs about AI vary based on the type of skills it assesses. Our focal prediction that people tend to perceive AI as less capable than humans in assessing interpersonal skills is supported by three main arguments. First, studies consistently

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show that people prefer services and decision-making provided by humans over AI^{9–11}. This algorithm aversion is particularly pronounced in subjective contexts that require AI to understand individual uniqueness (e.g., providing personalized medical advice)^{9,10,12} or emotion (e.g., recommending jokes or romantic partners)¹¹. Since interpersonal skills require an understanding of emotions and distinctive qualities of individuals¹³, people may relatively underestimate AI's capabilities to assess those skills. Second, studies have shown that people feel uneasy or threatened when AI exhibits interpersonal traits (e.g., emotionally expressive artificial agents)^{14,15}. Similarly, AI assessment of interpersonal skills may threaten people's identity, triggering ego-protective needs¹⁶ that are well-served by negative lay beliefs about AI selection technologies. Third, machines are often associated with coldness in popular culture, and this association may create implicit negative beliefs about AI's capability to assess interpersonal ("warm") skills. In a Pilot Study ($n = 102$) using the Implicit Association Test (IAT)¹⁷, we found that people hold implicit associations that AI is cold rather than warm. Because warmth is crucial in social relations¹⁸, lay beliefs that AI is less capable of assessing interpersonal skills may emerge.

This prediction carries important implications considering the long-standing skepticism surrounding human judgment in personnel selection. Human evaluations are subject to various biases and errors, influenced by rater (e.g., personality¹⁹) and ratee (e.g., race²⁰ and gender²¹) characteristics (see SI 3 for more examples). Such errors can account for more than half of the observed rating variance²². Although structured processes can improve human evaluations, problems persist, leading to unfair selection practices²³. AI technologies, however, offer promising solutions by analyzing applicants' resumes, job-seeking behaviors, and video interviews to identify suitable candidates^{24–26}. Notably, recent AI models can assess interpersonal traits, such as friendliness and personal qualities, by quantifying verbal and nonverbal behaviors (e.g., essays, facial expressions, prosodic information), often outperforming human evaluations (see SI 3 for more examples)^{26–34}.

To provide a test of AI's ability to evaluate interpersonal skills, we analyzed a Field Study dataset from a large hospital in South Korea ($n = 282$) that used both an AI assessment test (developed by MidasIT, a major provider of AI selection technology) and two human interviews (manager and executive interviews, see below) to evaluate applicants for nursing positions in late 2020 and 2021. For the manager interviews, senior nurses and HR staff evaluated applicants' nursing knowledge, motivation, and personality. In the executive interviews, a chief nursing officer and HR executives evaluated applicants' suitability for the role. We estimated a regression model with three predictors—the AI selection test score and the two human interview scores—predicting post-hire interpersonal performance evaluations by team leaders conducted in 2022. The evaluation score averaged six relevant items (e.g., work cooperation and communication; Cronbach's $\alpha = 0.92$; e.g., creates and maintains cooperative relationships with coworkers.; 5 = strongly agree). The AI score was the only significant predictor of post-hire interpersonal performance evaluations ($\beta = 0.10$, $SE = 0.04$, $t(267) = 2.45$, $P = 0.02$, 95% confidence interval (CI) [0.02, 0.18] vs. manager interview: $\beta = 0.07$, $SE = 0.04$, $t(267) = 1.56$, $P = 0.12$, 95% CI [−0.02, 0.15] and executive interview: $\beta = 0.08$, $SE = 0.04$, $t(267) = 1.84$, $P = 0.07$, 95% CI [−0.01, 0.16]; see SI 4 for details). We conducted a similar analysis using data from another hospital, again in the nursing context ($n = 107$). Findings further corroborated the effectiveness of AI selection technology ($\beta = 0.96$, $\text{Exp}(\beta) = 2.60$, $\text{Wald} = 2.71$, $P = 0.01$) than human interviewers ($\beta = 0.41$, $\text{Exp}(\beta) = 1.85$, $\text{Wald} = 1.32$, $P = 0.18$; see SI 2 for details). These findings collectively suggest preliminary evidence that AI technology is suitable for personnel selection tasks and, in particular, for the assessment of interpersonal skills, performing as well as or better than human experts.

While previous literature has shown prevalent algorithm aversion (for a review, see³⁵), recent research shows that people can also exhibit algorithm appreciation. For example, a study shows that people rely more on algorithms than on humans for estimates based on limited information (e.g., guessing an individual's weight from a photo)³⁶. Similarly, studies have found that people prefer or trust AI similarly to, or more than, humans in tasks such as behavior tracking³⁷, generating advertising content³⁸, and making decisions in areas including justice, health, and media³⁹. This raises the question of whether aversion to AI selection technologies varies depending on the specific skills they evaluate. Notably, previous literature suggests that people trust AI more when dealing with logical and objective tasks, such as giving financial advice (vs. dating advice)^{9,11}. AI recommenders are also rated as more competent for utilitarian goals (e.g., practical and objective use) than for hedonic goals (e.g., sensory pleasure and fun)⁴⁰. Therefore, people's negative lay beliefs may not generalize to assessments of analytical skills that are often associated with logic and computation, areas in which AI is widely perceived as competent and reliable.

We further argue that negative lay beliefs about AI assessment of interpersonal skills may have significant consequences on how people perceive coworkers selected by AI and how applicants behave during AI selection processes. First, we argue that these lay beliefs could lead managers to perceive AI-selected employees as having inferior interpersonal skills than those selected by humans. Since interpersonal skills are crucial for work across fields and professions⁴¹, managers' lay beliefs about AI capabilities may significantly impact manager-employee relationships and even influence performance evaluations^{41,42}. For instance, if AI-selected employees are presumed to have poor interpersonal skills, they may be overlooked for job assignments that require such skills. Second, we argue that applicants may reduce the emphasis on interpersonal skills during the selection process by AI versus humans. People's behavior tends to align with their beliefs^{43,44}. If applicants perceive AI as less capable than humans in assessing interpersonal skills, they should strategically prioritize other skills, such as analytical skills, during AI (vs. human) selection processes. Given that some AI selection technologies are designed to evaluate interpersonal traits (e.g., teamwork and friendliness)^{26,27} based on delivered messages from candidates, this behavioral adaptation might be damaging the applicant's prospects.

Finally, we argue that providing information about various AI techniques that assess interpersonal skills reduces negative lay beliefs. Misbeliefs may stem from insufficient knowledge, and can be reduced by providing accurate explanations⁴⁵. For instance, informing parents about their children's actual absent rates, often higher than their expectation, can correct false beliefs and subsequently decrease student absences⁴⁶. Similarly, providing explanations about how AI works (e.g., improving understanding of AI's healthcare diagnoses) could

enhance understanding and, consequently, willingness to adopt AI applications^{10,35}. Therefore, we predict that informing individuals about various examples of recently developed AI selection technologies that can assess interpersonal skills may increase knowledge, and ultimately help reduce negative lay beliefs.

To summarize our predictions, we propose that (a) people tend to perceive AI as less capable than humans in assessing applicants' interpersonal skills. We further propose that such lay beliefs (b) have consequences for how others perceive AI-selected individuals, (c) have consequences for the kind of skills that applicants emphasize in AI selection processes, (d) do not extend to lay beliefs about AI assessment of analytical skills, and (e) can be reduced by informing people about various advanced AI technologies that can assess interpersonal skills. We test these predictions in six studies. In Studies 1 to 3, we measure lay beliefs through a survey and experiments. Studies 4 and 5 demonstrate the potential consequences of such lay beliefs on managers' decisions and job candidates' interviewing strategies. Finally, Study 6 shows that providing information about various AI techniques that assess interpersonal skills reduces negative lay beliefs.

Results

Studies 1–3: Lay beliefs about AI assessing interpersonal skills

Studies 1 to 3 find lay beliefs about AI assessing applicants' interpersonal skills. Study 1 tested which pieces of information people tend to check when asked to guess whether the applicant had passed the interview assessed by either an AI or human, from four assessment scores (including interpersonal and analytical skills assessment scores). Participants ($n = 200$) could only choose to reveal one of the four assessment scores and were informed that a bonus would be paid for a correct guess (i.e., incentive-compatible design). Individuals tend to rely on and give more weight to information that aligns with their beliefs^{43,44}. Therefore, if individuals doubt AI's capabilities to assess interpersonal skills, they may rely on what they perceive as more reliable information. Consequently, participants in the AI (versus human) condition would be less likely to seek interpersonal skills assessment scores than analytical skills assessment scores.

As pre-registered, we compared the proportion of participants who chose to view the interpersonal skills assessment score versus the analytical skill assessment score as a function of assessment agent type. As predicted, the assessment agent type significantly influenced the decision of which information to view ($\chi^2(1) = 19.28$, $P < 0.001$). In the AI condition, 42.7% of participants chose the interpersonal skills score, while 75.3% did in the human condition. The odds of choosing to view interpersonal skills over analytical skills assessment scores were 4.09 times higher (95% CI[2.15, 7.77]) in the human than in the AI condition.

Study 2 tested lay beliefs through another experiment. Participants ($n = 201$) were randomly assigned to one of two selection processes assessed by either AI or humans. Participants in the AI [Human] condition read a brief description of AI interviewers [HR personnel] and rated the capabilities of the AI interviewers [HR personnel] in assessing job candidates' interpersonal and analytical skills, respectively (e.g., AI interviewers would be capable of evaluating the job candidates' interpersonal skills.; 7-point Likert scale; 7 = strongly agree). We expected participants to show less trust towards AI (vs. humans) in assessing interpersonal skills but not analytical skills.

We examined the perceived capabilities of the AI versus human (i.e., assessment agent; between-participants factor) in assessing interpersonal and analytical skills (i.e., skill type; within-participants factor) using a mixed ANOVA. Our hypothesis was supported by the significant interaction effect ($F(1, 197) = 93.86$, $P < 0.001$, $\eta^2 = 0.323$), showing a lower perceived capability of AI (vs. human) for evaluating interpersonal skills but not analytical skills (Fig. 1). Specifically, planned contrast revealed that participants perceived the AI as less capable in the assessment of interpersonal skills ($M = 3.28$, $SD = 1.77$) compared to humans ($M = 5.24$, $SD = 1.30$; $F(1, 197) = 80.07$, $P < 0.001$, $\eta^2 = 0.289$). However, participants perceived that AI's capabilities in assessing analytical skills ($M = 4.86$, $SD = 1.60$) were comparable to that of humans ($M = 4.64$, $SD = 1.41$; $F(1, 197) = 1.10$, $P = 0.30$, $\eta^2 = 0.006$).

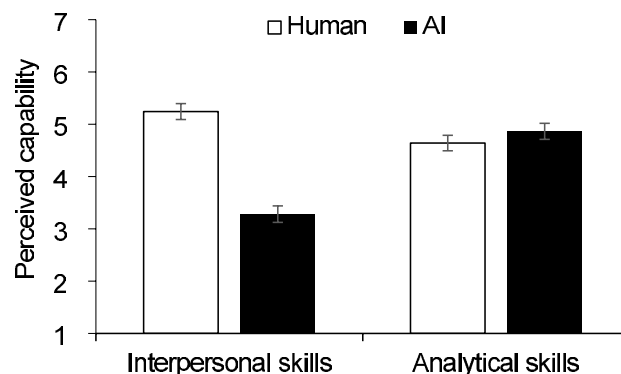


Fig. 1. Perceived capabilities as a function of selection agent and skill type in Study 2. Note: Error bars indicate 1 standard error of the mean.

We replicated Study 2 ($n = 63$) using a similar design but replacing the term AI interviewer with AI assessment tools designed to analyze candidates' responses to job- and personality-related questions and their performance in mini-games^{2,3}. The results were consistent with our earlier findings (see SI 5 for details).

Study 3 ($n = 343$) explored lay beliefs about AI selection with practitioners by directly asking managers to compare their perceptions about AI and human assessments. We used surveys collected by MidasIT, a major provider of AI selection technology in Korea. Of the participants, 81.6% ($n = 280$) were responsible for hiring tasks. We focused on two questions asking whether they believed AI or humans were better at predicting an applicant's interpersonal and analytical skills, respectively (1⁷ indicated definitely AI [human] interviews). We expected managers to believe AI is less effective than humans in predicting an applicant's interpersonal skills but not analytical skills.

As predicted, managers rated AI as less capable than humans in predicting interpersonal skills ($M = 5.02$, $SD = 1.37$; one-sample t -test comparing against 4: $t(342) = 13.81$, $P < 0.001$, Cohen's d (d) = 0.75, 95% CI[0.88, 1.17]), but as more capable in predicting analytical skills ($M = 3.44$, $SD = 1.61$; one-sample t -test: $t(342) = -6.47$, $P < 0.001$, $d = -0.35$, 95% CI[-0.73, -0.39]). Next, we estimated a linear mixed model with a random intercept and slope for each manager, where we regressed managers' lay beliefs on each of the two assessed skill types while controlling for their involvement in hiring tasks (i.e., by adding fixed effects for a dummy variable indicating whether each respondent was in charge of hiring tasks or not). The results showed the same pattern of significance ($b = 1.58$, $t(342) = 14.88$, $P < 0.001$, 95% CI[1.37, 1.58]).

Studies 4–5: Consequences of lay beliefs about AI assessing interpersonal skills

The previous studies suggest people negatively perceive AI's capabilities in assessing applicants' interpersonal (vs. analytical) skills. Studies 4 and 5 examine the potential consequences of these lay beliefs for office workers and job applicants. Study 4 tested whether office workers hold biases towards AI-selected employees as opposed to those selected by humans. If office workers are skeptical of AI's capabilities in assessing interpersonal skills, they might also perceive AI-selected employees as having inferior interpersonal skills than those selected by humans. As a result, we predict that office workers would be less likely to assign tasks requiring interpersonal skills to AI-selected employees (vs. human-selected), but this bias would not extend to tasks requiring analytical skills. That is, we predict that office workers would not assign tasks requiring analytical skills more to employees selected by humans than AI.

We recruited 102 office workers in large Korean companies, which are increasingly using AI assessments. From the managers' perspective, the office workers evaluated two hypothetical employees, one hired through a conventional HR interview and the other hired through an AI interview. Participants were then asked to indicate which employee they would assign tasks requiring interpersonal skills and analytical skills, respectively (i.e., using a 7-point Likert scale where 1⁷ indicated a definite preference for the HR [AI] interview hired employee).

As predicted, the office workers indicated that they would assign tasks requiring interpersonal skills less to the employee hired through the AI interview than the one hired through the human interview ($M = 2.55$, $SD = 1.36$; one sample t -test with a comparison against 4: $t(101) = -10.76$, $P < 0.001$, $d = -1.07$, 95% CI[-1.72, -1.18]). However, they were more likely to assign tasks requiring analytical skills to the AI-interviewed employee ($M = 5.29$, $SD = 1.41$; one sample t -test: $t(101) = 9.26$, $P < 0.001$, $d = 0.92$, 95% CI[1.02, 1.57]).

Study 5 tested another potential consequence of the lay beliefs, focusing on applicants' strategies during the selection process assessed by AI versus humans. We investigated which type of skills is considered important for each of the two selection types. Given that people's behavior tends to reflect their beliefs^{43,44}, if job applicants are skeptical about AI's capabilities in assessing interpersonal skills, applicants may infer that showcasing these skills will have less impact on the selection decision. Consequently, they should emphasize their interpersonal skills less in the AI-assessed selection process than in the human-assessed one. Conversely, we anticipated no such difference for analytical skills.

We recruited 107 participants from South Korea, who had experienced a selection process evaluated by both AI and humans. We asked participants to recall their most recent interviews of each type and then to indicate in which of the two interviews they had emphasized their interpersonal and analytical skills relatively more (where 1⁷ indicated they definitely emphasized the skill more in the human [AI] interview; the order of the two skill questions was counterbalanced, which has no significant effects on participants' responses (all p -values > 0.4 ; SI 6)).

As predicted, participants emphasized interpersonal skills less during the AI interview than human interview ($M = 3.31$, $SD = 1.67$; one sample t -test against 4: $t(106) = -4.28$, $P < 0.001$, $d = -0.41$, 95% CI[-1.01, -0.37]). On the other hand, participants emphasized analytical skills more during the AI interview than human interview ($M = 4.42$, $SD = 1.51$; one sample t -test against 4: $t(106) = 2.88$, $P = 0.01$, $d = 0.28$, 95% CI[0.13, 0.71]).

Study 6: Alleviating negative lay beliefs about AI assessing interpersonal skills

The previous studies highlight negative lay beliefs about AI that could cause problems in organizational settings, affecting managers' task assignments and job applicants' strategies during selection processes. However, our field data analysis suggests that AI selection technology can outperform humans in predicting interpersonal performances. Likewise, research continues to demonstrate the efficacy of AI in measuring interpersonal traits, such as friendliness^{26–29}. Therefore, Study 6 investigated a strategy to reduce the negative lay beliefs about AI assessing interpersonal skills. We speculated that people may find it challenging to grasp the concepts of AI technologies being used to assess interpersonal skills. Therefore, we predicted that informing people of various examples of recently developed AI selection technologies that can assess interpersonal skills would help reduce the negative lay beliefs.

To test this, we recruited 231 participants who had heard of AI selection technologies to examine the change in consumers' existing beliefs toward AI's capabilities. We told participants that they would take on two

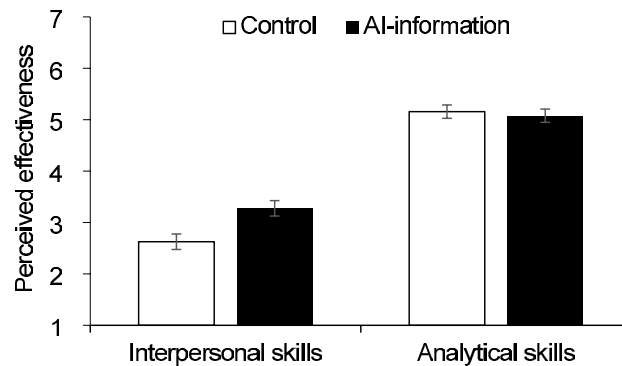


Fig. 2. Perceived capabilities of AI as a function of experimental condition and skill type in Study 6. Note: Error bars indicate 1 standard error of the mean.

supposedly unrelated tasks. In the first task, participants were randomly assigned to one of the two conditions—AI-information vs. control condition. Participants in the AI-information condition summarized a news article about the trend of AI selection technologies that capture facial expressions, emotions, social skills, and more, while those in the control condition summarized a news article on the use of AI in separate contexts. For the second task, participants were asked to evaluate which type of the two selection processes would be more effective in selecting applicants with outstanding interpersonal and analytical skills, respectively (where 1⁷ represents definitely more effective in human [AI] interview).

As in Study 2, participants in the control condition thought that AI was less effective than humans in selecting applicants with outstanding interpersonal skills ($M = 2.63$, $SD = 1.56$; one sample t -test: $t(114) = -9.42$, $P < 0.001$, $d = -0.88$, 95% $CI[-1.66, -1.08]$); however, they thought AI was more effective for selecting applicants with outstanding analytical skills ($M = 5.16$, $SD = 1.47$; $t(114) = 8.46$, $P < 0.001$, $d = 0.79$, 95% $CI[0.89, 1.43]$). Using a mixed ANOVA, we then tested whether informing participants about advanced AI selection technologies would increase their perception of AI (vs. humans) in assessing interpersonal skills, but not in terms of analytical skills (Fig. 2). As expected, the interaction was significant ($F(1, 229) = 7.67$, $P = 0.01$, $\eta^2 = 0.032$). A further breakdown of this interaction showed that the perceived capabilities of AI assessing interpersonal skills were higher in the AI-information condition ($M = 3.28$, $SD = 1.67$) than in the control condition ($M = 2.63$, $SD = 1.56$; $F(1, 229) = 9.34$, $P = 0.03$, $\eta^2 = 0.039$). By contrast, the perceived effectiveness of AI in assessing analytical skills did not differ significantly between the two conditions ($M_{AI-information} = 5.16$, $SD = 1.47$ vs. $M_{control} = 5.08$, $SD = 1.32$; $F(1, 229) = 0.19$, $P = 0.67$, $\eta^2 = 0.01$).

Discussion

We delineate lay beliefs by differentiating AI assessments for interpersonal and analytical skills, which correspond to the two major dimensions of human judgment (i.e., warmth and competence, respectively)^{47,48}. Our results document prevailing lay beliefs among hiring managers and labor market participants that AI for personnel selection is less capable of assessing interpersonal skills, but not analytical skills, than humans. These beliefs can bias perspectives, impacting both managers and job applicants. We found that managers may be less inclined to assign tasks requiring interpersonal skills to employees selected through AI (vs. humans) and job candidates may downplay their interpersonal skills (vs. analytical ones) during AI selection processes. Notably, the final study showed that providing information about advanced AI techniques that assess interpersonal skills can reduce such negative lay beliefs.

Our research has significant theoretical implications. First, it advances our understanding of perceptions toward AI in personnel selection. Prior research in this area has mainly focused on how individual traits or aspects of AI adoption affect perceptions of fairness and efficiency^{38,49,50}. For example, perceptions of unfairness increase when individuals are unfamiliar with AI or when AI is introduced in the later stages of the selection process⁵¹. In contrast, our study expands this area by investigating how people perceive AI's capabilities to assess distinct skills, particularly focusing on interpersonal skills (vs. analytical skills). Specifically, our findings indicate that negative perceptions are not universally consistent across all facets of AI's capabilities—people perceive AI as less effective in assessing candidates' interpersonal skills, but not analytical skills. This highlights the importance of distinguishing between various skills when studying AI perception in personnel selection.

Second, our research contributes to the existing literature on algorithm aversion and algorithm appreciation by demonstrating that algorithm aversion in the domain of personnel selection is complex and task-dependent¹¹. Previous studies show a general aversion towards AI in various domains³⁵, and this trend is similarly observed in the context of AI's application in personnel selection^{50,51}. Our findings, however, show that while there is a tendency for negative perceptions toward AI in assessing interpersonal skills, there simultaneously exists a preference for AI in the assessment of analytical skills. This suggests that within a single domain, such as AI

in personnel selection, algorithm aversion and algorithm appreciation can coexist and vary depending on the nature of the task at hand.

Our findings also have important practical implications. We first suggest that lay beliefs about AI's competency could hinder its successful adoption in interview processes, especially as most jobs require substantial interpersonal engagement⁴¹. Considering our Field Study results, reducing negative beliefs among HR professionals could help improve selection outcomes. Companies such as MidasIT, Eightfold, Findem, HireVue, hireEZ, and Pymetrics promote their AI selection software primarily in terms of diverse assessment methods and efficiency in talent sourcing⁵². However, our research suggests that firms should also emphasize the AI assessment tool's capabilities in assessing interpersonal skills, as doing so can reduce negative lay beliefs about AI (Study 6) and enhance its successful adoption.

We also suggest these lay beliefs may affect workplace dynamics and talent selection processes. While prior studies have focused primarily on immediate perceptions of AI selection technologies (e.g., trust, fairness)^{47,51,53,54}, relatively few have examined the potential consequences it may bring. We provide initial evidence that managers might perceive AI-selected (vs. human-selected) employees as less competent in roles that require interpersonal skills, such as sales, service, or leadership roles, but not in roles that are heavily analytical, such as accounting or data analysis. Concurrently, job applicants might inadvertently undermine their own employment prospects during AI-based selection processes by placing less emphasis on interpersonal skills such as social competency and leadership experience.

While our studies capture general lay beliefs across a broad population, algorithm aversion is context-dependent^{9–12}. Therefore, lay beliefs may vary depending on contextual factors, such as perceived hiring task complexity, AI expertise, or job type (e.g., fewer interpersonal demands). Furthermore, while our Field Study findings leverage real-world data to enhance external validity, they are limited by particular situations, differences in candidate pools, and the varying amounts of information collected by AI compared to humans (see SI 4 for details)⁵⁵. Future research should address these contextual and methodological gaps to better understand the effectiveness of AI (vs. humans) and lay beliefs about its use in personnel selection.

Future research could also probe other potential lay beliefs. For example, in Studies 2, 4, and 6, we additionally examined people's perceptions of AI's capabilities to predict employee turnover. We found that people tend to perceive AI as less capable than humans at predicting employee turnover (Study 2: $M_{AI} = 3.44$, $SD = 1.58$ vs. $M_{human} = 4.15$, $SD = 1.55$; $t(197) = 3.19$, $P = 0.02$, $d = 0.45$, 95% CI[0.27, 1.15]; Study 4: $M = 4.34$, $SD = 1.84$; $t(101) = 1.88$, $P = 0.06$, $d = 0.19$, 95% CI[−0.02, 0.70], compared to a midpoint of 4 representing the equal perceived capability of AI and humans; Study 6: $M = 3.34$, $SD = 1.66$; $t(230) = -6.04$, $P < 0.001$, $d = -0.40$, 95% CI[−0.88, −0.45]). These findings suggest that managers may hold the unfounded assumption that AI-selected employees are more likely to quit, potentially harming manager-employee relationships.

Methods

Participants

For Studies 1 and 2, participants were collected from Amazon's Mechanical Turk and Prolific, respectively. Study 3 survey responses were collected by MidasIT. For Studies 4 to 6, participants were recruited by market research agencies. Except for Study 3, which had a sample size of 343 collected by the company, the other studies aimed to recruit 100 participants per cell. We determined the required sample size by conducting a power analysis using G*Power to achieve 80% power with a Type I error rate (α) of 5%, targeting small to medium main effects (i.e., Cramér's $V = 0.2$, $f(V) = 0.2$, $d = 0.35$). Post hoc power analyses, with a target alpha level of 0.05, confirmed that the achieved sample sizes provided sufficient power (i.e., ≥ 0.78) across all studies to detect small to medium effects. Four studies (Studies 1, 2, 4, and 5) were preregistered. All studies' materials, data, output results, and preregistration documents are available in OSF (osf.io/mgpb83/?view_only=b5dc0f91bf3545f38b10c2143f2d7d19), except for the field studies data, which are restricted due to non-disclosure agreements.

Design and procedure

Study 1 employed a 2 (assessment type: AI vs. human; between-participants) x 2 (skill type: interpersonal vs. analytical skills; within-participants) mixed factorial design. Study 2 employed a single-factor (AI vs. human) between-participants design. Studies 3–5 employed a survey. Study 6 employed a 2 (article type: AI-information vs. control condition; between-participants) x 2 (skill type: interpersonal vs. analytical skills; within-participants) mixed factorial design. Lay beliefs are commonly measured directly through self-reported perceptions, where participants explicitly indicate their subjective beliefs in a given domain^{56,57}, and indirectly by assessing behavioral tendencies or decision-making patterns that reflect those beliefs^{56,58}. Accordingly, we operationalize lay beliefs about AI-based selection tools using both direct measures (i.e. by directly asking participants to rate AI's capability in assessing candidates' interpersonal skills in Studies 2, 3, and 6) and indirect measures (i.e., by assessing whether participants rely more on candidates' interpersonal or analytical scores to determine if a candidate passed an AI assessment in Study 1). Detailed design and procedure are presented in SI 6.

Data availability

All studies' materials, data, output results, and preregistration documents are available in OSF (https://osf.io/mgpb83/?view_only=b5dc0f91bf3545f38b10c2143f2d7d19), except for the field studies data, which are restricted due to non-disclosure agreements.

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Author contributions

I.C., Y.H., and S.P. designed the study and I.C. conducted the experiments and performed the analysis. Y.H. and S.P. supervised the project. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

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